

FEATURES FORMED BY WATER

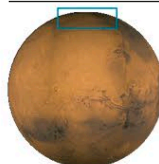
Both liquid and solid water have formed and shaped surface features on Mars. Giant channel-like valleys emerge fully formed out of the landscape. Some of these were cut during catastrophic floods, others were formed by water flowing more gradually through networks of river valleys, and others still were carved by glaciers. Some features suggest Mars once had seas, although the evidence is inconclusive. However, any potential rivers and seas have long since vanished, and only water ice remains, most markedly in the two ice plateaus that cap the planet.

REULL VALLIS

Long, wide river channels are etched into the surface, revealing that huge volumes of water flowed across Mars billions of years ago.

PLANUM BOREUM

North Polar Region



TYPE Polar ice cap

AGE Under 2.5 billion years

DIAMETER 685 miles (1,100 km)

Two bright white polar caps stand out against the otherwise dark surface of Mars. The one roughly centered on the north pole is officially named Planum Boreum—the Northern Plain—although it is generally referred to as the North Polar Cap. Both this and its southern counterpart are easy to detect from Earth, but spacecraft have also flown over the poles, allowing monitoring of daily, seasonal, and longer-term change.

The North Polar Cap on Mars is an ice-dominated mound that stands several miles above the surrounding terrain. It consists of a virtually permanent cap of water ice, which is either covered by or free of a deposit of carbon-dioxide ice, depending on the time of the Martian year. The cap is roughly circular but—as is also the case for the South Polar Cap—its bright ice forms a distinctive swirling, loosely spiral pattern when seen from above (see p.163).

The entire region is in darkness for about 6 months during the Martian winter.

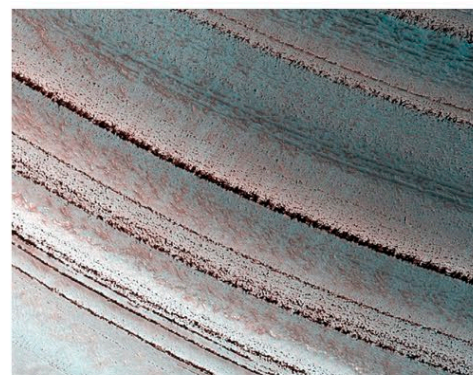


POLAR POLYGONS

Polygon-shaped structures, similar to those found in Earth's polar regions, pattern parts of Mars's polar landscape. On Earth, they form as a result of stresses induced by repeated freezing and thawing of water.

This is when carbon dioxide in the atmosphere condenses into frost and snow and not only covers the water-ice cap, but also the surrounding region, down to latitudes of about 65° north. When spring turns to summer and the Sun is permanently in the polar sky, its warmth evaporates the carbon dioxide and turns some of the water ice directly into vapor. The polar cap shrinks until just water ice remains.

The cap is not made exclusively of ice but consists of layers of ice and layers of dusty sediment. Frost grains form around small particles of dust during winter dust storms in much the same way that hailstones form on Earth. These cover the ground until the frost is evaporated in the warmer months, leaving a layer of dust. The yards-deep layers take



LAYERED DEPOSITS

Layers of ice at the Martian north pole attest to past variations in the planet's climate. The layers are exposed at the edge of the ice sheet, which slopes downhill from the bottom to the top in this image. The thickness of the ice is about 0.6 miles (1 km).

millions of years to form, building up at the rate of about 0.04 in (1 mm) per year. A study of these layers will reveal the history of the Martian climate.

CLIFFS NEAR THE NORTH POLE

This close-up view of the Martian North Polar Cap shows water ice close to cliffs about 1.2 miles (2 km) high. Dark material in the calderalike structures and dune fields could be volcanic ash.